

UNIT V

NETWORKING ESSENTIALS

1.1 NETWORKING COMPONENTS AND SERVICES

Data Communications

Data communications are the exchange of data between two devices via some form of transmission medium such as a wire cable. For data communications to occur, the communicating devices must be part of a communication system made up of a combination of hardware (physical equipment) and software (programs).

The effectiveness of a data communications system depends on four fundamental characteristics

1. Delivery: The system must deliver data to the correct destination. Data must be received by the intended device or user and only by that device or user.

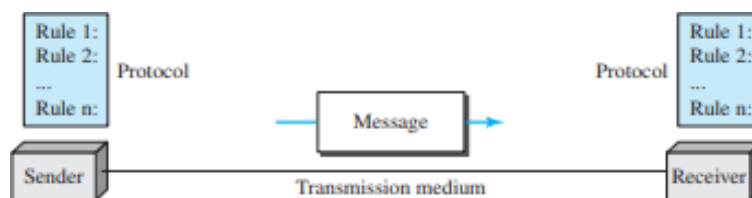
2. Accuracy: The system must deliver the data accurately. Data that have been altered in transmission and left uncorrected are unusable.

3. Timeliness: The system must deliver data in a timely manner

4. Jitter: Jitter refers to the variation in the packet arrival time. It is the uneven delay in the delivery of audio or video packets.

Data Communication Components

A data communications system has five components as shown in the below figure.



1. Message. The message is the information (data) to be communicated. Popular forms of information include text, numbers, pictures, audio, and video.

2. Sender. The sender is the device that sends the data message. It can be a computer, workstation, telephone handset, video camera, and so on.

3. Receiver. The receiver is the device that receives the message. It can be a computer, workstation, telephone handset, television, and so on.

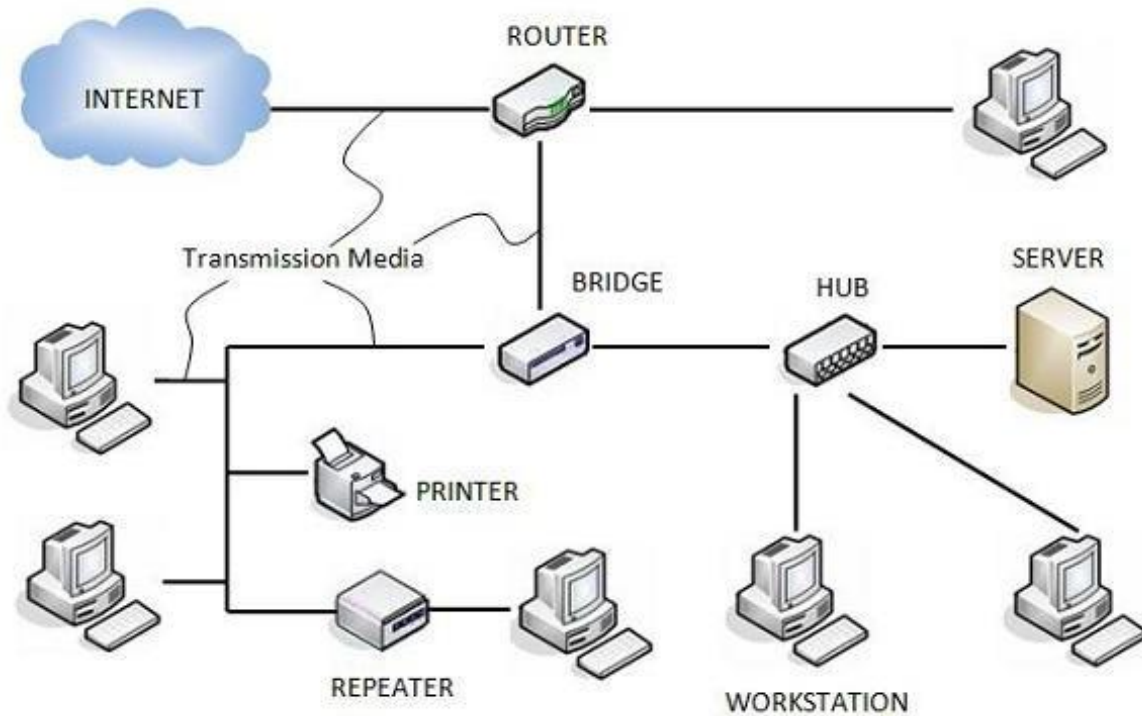
4. Transmission medium. The transmission medium is the physical path by which a message travels from sender to receiver. Some examples of transmission media include twisted-pair wire, coaxial cable, fiber-optic cable, and radio waves.

5. Protocol. A protocol is a set of rules that govern data communications. It represents an

agreement between the communicating devices. Without a protocol, two devices may be connected but not communicating, just as a person speaking French cannot be understood by a person who speaks only Japanese.

Computer Network Hardware Components

The below figure shows a network along with its components.



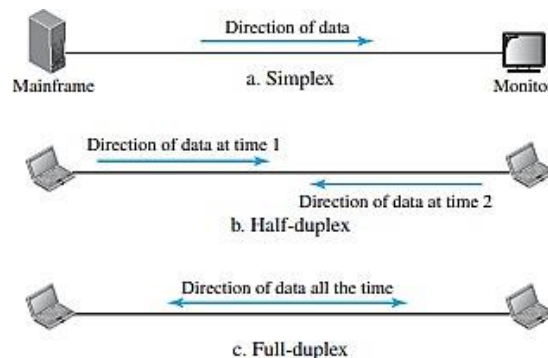
1. **Servers** – Servers are high-configuration computers that manage the resources of the network. The network operating system is typically installed in the server and so they give user accesses to the network resources. Servers can be of various kinds: file servers, database servers, print servers etc.
2. **Clients** – Clients are computers that request and receive service from the servers to access and use the network resources.
3. **Peers** – Peers are computers that provide as well as receive services from other peers in a workgroup network.
4. **Transmission Media** – Transmission media are the channels through which data is transferred from one device to another in a network. Transmission media may be guided media like coaxial cable, fibre optic cables etc; or maybe unguided media like microwaves, infra-red waves etc.
5. **Connecting Devices** – Connecting devices act as middleware between networks or computers, by binding the network media together. Some of the common connecting devices are: Routers, Bridges, Hubs, Repeaters, Gateways and Switches.

Computer Network Software Components

1. **Networking Operating System** – Network Operating Systems is typically installed in the server and facilitate workstations in a network to share files, database, applications, printers etc.
2. **Protocol Suite** – A protocol is a rule or guideline followed by each computer for data communication. Protocol suite is a set of related protocols that are laid down for computer networks. The two popular protocol suites are –
 - a. OSI Model (Open System Interconnections)
 - b. TCP / IP Model

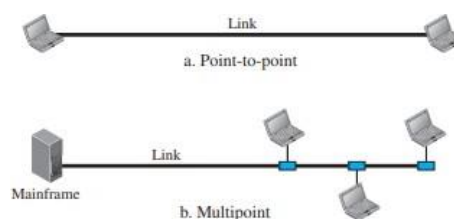
Data Flow

Communication between two devices can be simplex, half-duplex, or full-duplex as shown in below Figure.



1. **Simplex:** In simplex mode, the communication is unidirectional, as on a one-way street. Only one of the two devices on a link can transmit; the other can only receive. **Keyboards and traditional monitors** are examples of simplex devices.
2. **Half-Duplex** In half-duplex mode, each station can both transmit and receive, but not at the same time. When one device is sending, the other can only receive, and vice versa. **Walkie-talkies and CB (citizens band) radios** are both half-duplex systems.
3. **Full-Duplex** In full-duplex mode (also called duplex), both stations can transmit and receive simultaneously. One common example of full-duplex communication is the **telephone network**.

Types of Connection in Networking



1. **Point-to-Point:** A point-to-point connection provides a dedicated link between two devices. The entire capacity of the link is reserved for transmission between those two devices. Most point-to-point connections use an actual length of wire or cable to connect the two ends, but other options, such as microwave or satellite links
2. **Multipoint:** A multipoint (also called multidrop) connection is one in which more than two specific devices share a single link. In a multipoint environment, the capacity of the channel is shared, either spatially or temporally. If several devices can use the link simultaneously, it is a spatially shared connection. If users must take turns, it is a timeshared connection.

Physical Topology

The term physical topology refers to the way in which a network is laid out physically. Two or more devices connect to a link; two or more links form a topology. The topology of a network is the geometric representation of the relationship of all the links and linking devices (usually called nodes) to one another.

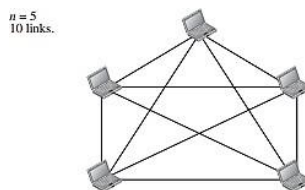
1. Mesh Topology:

In a mesh topology, every device has a dedicated point-to-point link to every other device. The term dedicated means that the link carries traffic only between the two devices it connects.

A mesh offers several advantages over other network topologies. First, the use of dedicated links guarantees that each connection can carry its own data load, thus eliminating the traffic problems that can occur when links must be shared by multiple devices.

Second, a mesh topology is robust. If one link becomes unusable, it does not incapacitate the entire system.

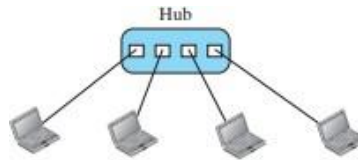
Third, there is the advantage of privacy or security. When every message travels along a dedicated line, only the intended recipient sees it.



2. Star Topology

In a star topology, each device has a dedicated point-to-point link only to a central controller, usually called a hub. The devices are not directly linked to one another. Unlike a mesh topology, a star topology does not allow direct traffic between devices.

The controller acts as an exchange: If one device wants to send data to another, it sends the data to the controller, which then relays the data to the other connected device



In a star, each device needs only one link and one I/O port to connect it to any number of others. This factor also makes it easy to install and reconfigure.

Robustness. If one link fails, only that link is affected. All other links remain active. This factor also lends itself to easy fault identification and fault isolation.

3. Bus Topology

A bus topology, on the other hand, is multipoint. One long cable acts as a backbone to link all the devices in a network. Nodes are connected to the bus cable by drop lines and taps. A drop line is a connection running between the device and the main cable. A tap is a connector that either splices into the main cable or punctures the sheathing of a cable to create a contact with the metallic core.

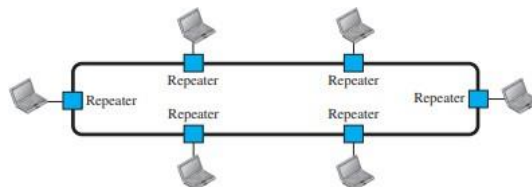


Advantages of a bus topology include ease of installation. Backbone cable can be laid along the most efficient path, then connected to the nodes by drop lines of various lengths.

In a bus, this redundancy is eliminated. Only the backbone cable stretches through the entire facility. Each drop line has to reach only as far as the nearest point on the backbone.

4. Ring Topology

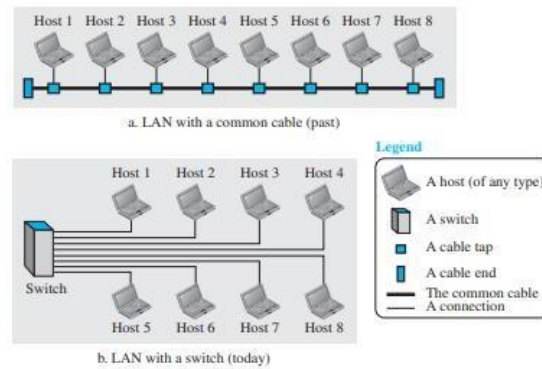
In a ring topology, each device has a dedicated point-to-point connection with only the two devices on either side of it. A signal is passed along the ring in one direction, from device to device, until it reaches its destination.



Network Types

1. Local Area Network

A local area network (LAN) is usually privately owned and connects some hosts in a single office, building, or campus. Depending on the needs of an organization, a LAN can be as simple as two PCs and a printer in someone's home office, or it can extend throughout a company and include audio and video devices.



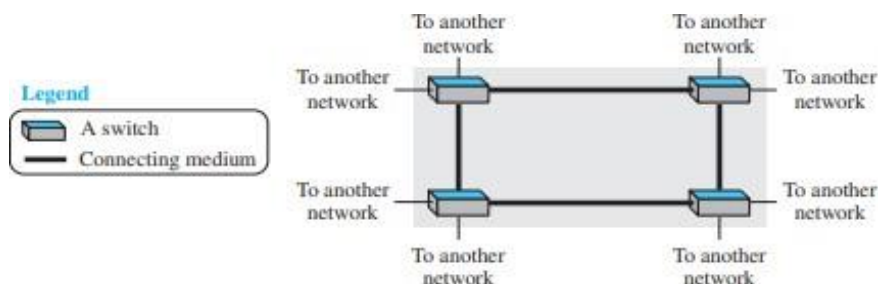
2. Wide Area Network

A wide area network (WAN) is also an interconnection of devices capable of communication. However, there are some differences between a LAN and a WAN. A LAN is normally limited in size, spanning an office, a building, or a campus; a WAN has a wider geographical span, spanning a town, a state, a country, or even the world.

Point-to-Point WAN A point-to-point WAN is a network that connects two communicating devices through a transmission media (cable or air).



Switched WAN A switched WAN is a network with more than two ends. A switched WAN, as we will see shortly, is used in the backbone of global communication today.



1.2 IP ADDRESSING

An IP address, or Internet Protocol address, is a unique string of numbers assigned to each device connected to a computer network that uses the Internet Protocol for communication. It serves as an identifier that allows devices to send and receive data over the network, ensuring that this data reaches the correct destination. **Working of IP addresses**

The working of IP addresses is similar to other languages. It can also use some set of rules to send information. Using these protocols we can easily send, and receive data or files to the connected devices. There are several steps behind the scenes. Let us look at them

- Your device directly requests your Internet Service Provider which then grants your device access to the web.
- And an IP Address is assigned to your device from the given range available.
- Your internet activity goes through your service provider, and they route it back to you, using your IP address.
- Your IP address can change. For example, turning your router on or off can change your IP Address.
- When you are out from your home location your home IP address doesn't accompany you. It changes as you change the network of your device.

Classification of IP Address Based on Addressing Scheme

IPv4:

This is the most common form of IP Address. It consists of four sets of numbers separated by dots. For example, 192.158.1.38. Each set of numbers can range from 0 to 255. This format can support over 4 billion unique addresses. Here's how the structure is broken down:

Four Octets: Each octet represents eight bits, or a byte, and can take a value from 0 to 255. This range is derived from the possible combinations of eight bits ($2^8 = 256$ combinations).

Example of IPv4 Address: 192.168.1.1

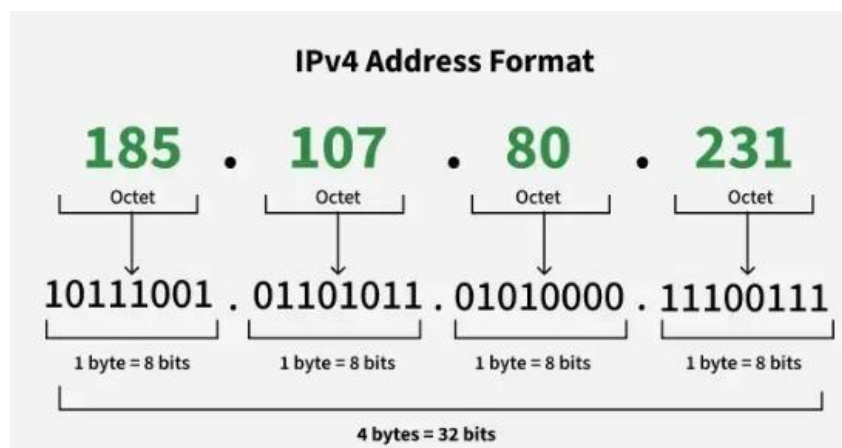
192 is the first octet

168 is the second octet

1 is the third octet

1 is the fourth octet

Each part of the IP address can indicate various aspects of the network configuration, from the network itself to the specific device within that network.



IPv6:

IPv6 addresses were created to deal with the shortage of IPv4 addresses. They use 128 bits instead of 32, offering a vastly greater number of possible addresses. These addresses are expressed as eight groups of four hexadecimal digits, each group representing 16 bits. The groups are separated by colons.

Example of IPv6 Address: 2001:0db8:85a3:0000:0000:8a2e:0370:7334

Each group (like **2001**, **0db8**, **85a3**, etc.) represents a 16-bit block of the address.

Classification of IP Address Based on Usage**Public IP Addresses**

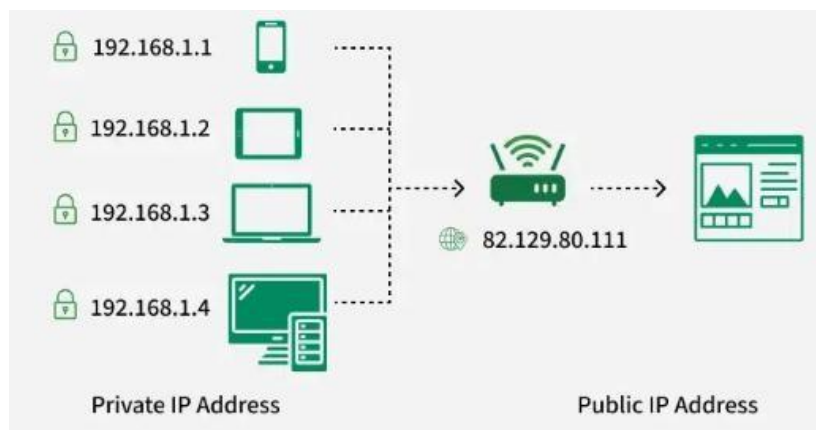
A Public IP address is assigned to every device that directly accesses the internet. This address is unique across the entire internet. Here are the key characteristics and uses of public IP addresses:

Uniqueness: Each public IP address is globally unique. No two devices on the internet can have the same public IP address at the same time.

Accessibility: Devices with a public IP address can be accessed directly from anywhere on the internet, assuming no firewall or security settings block the access.

Assigned by ISPs: Public IP addresses are assigned by Internet Service Providers (ISPs). When you connect to the internet through an ISP, your device or router receives a public IP address.

Types: Public IP addresses can be static (permanently assigned to a device) or dynamic (temporarily assigned and can change over time).

**Private IP Addresses**

Private IP addresses are used within private networks (such as home networks, office networks, etc.) and are not routable on the internet. This means that devices with private IP addresses cannot directly communicate with devices on the internet without a translating mechanism like a router performing Network Address Translation (NAT). Key features include:

- **Not globally unique:** Private IP addresses are only required to be unique within their own network. Different private networks can use the same range of IP addresses without conflict.
- **Local communication:** These addresses are used for communication between devices within the same network. They cannot be used to communicate directly with devices on the internet.
- **Defined ranges:** The Internet Assigned Numbers Authority (IANA) has reserved specific IP address ranges for private use:
- **IPv4:** 10.0.0.0 to 10.255.255.255, 172.16.0.0 to 172.31.255.255, 192.168.0.0 to 192.168.255.255
- **IPv6:** Addresses starting with FD or FC

Classification of IP Address Based on Assignment Method

Static IP Addresses:

- These are permanently assigned to a device, typically important for servers or devices that need a constant address.
- Reliable for network services that require regular access such as websites, remote management.

Dynamic IP Addresses:

- Temporarily assigned from a pool of available addresses by the Dynamic Host Configuration Protocol (DHCP).
- Cost-effective and efficient for providers, perfect for consumer devices that do not require permanent addresses.

Classes of IPv4 Address

There are around 4.3 billion IPv4 addresses and managing all those addresses without any classification is next to impossible. For easier management and assignment IP addresses are organized in numeric order and divided into the following 5 classes:

- **Class A (1.0.0.0 to 127.255.255.255):**
 - Used for very large networks (like multinational companies).
 - Supports up to 16 million hosts per network.
 - Example: 10.0.0.1 (Private IP in this class).
- **Class B (128.0.0.0 to 191.255.255.255):**
 - Used for medium-sized networks, such as large organizations.
 - Supports up to 65,000 hosts per network. Example: 172.16.0.1 (Private IP class)

- **Class C (192.0.0.0 to 223.255.255.255):**
 - Used for smaller networks, like small businesses or home networks.
 - Supports up to 254 hosts per network.
 - Example: 192.168.1.1 (Private IP in this class).
- **Class D (224.0.0.0 to 239.255.255.255):**
 - Reserved for multicast groups (used to send data to multiple devices at once).
 - Not used for traditional devices or networks.
- **Class E (240.0.0.0 to 255.255.255.255):**
 - Reserved for experimental purposes and future use.

IP Class	Address Range	Maximum number of networks
Class A	1-126	126 ($2^7 - 2$)
Class B	128-191	16384
Class C	192-223	2097152
Class D	224-239	Reserve for multitasking
Class E	240-254	Reserved for Research and development

1.3 CONFIGURING AND MANAGING THE CAMPUS NETWORK

A campus network encompasses interconnected LANs spread across multiple buildings or locations within a defined geographical area. These networks are structured to ensure seamless communication between devices, with a focus on reliability, efficiency, and scalability. This topic explores the key components, configurations, and management strategies essential for building and maintaining an efficient campus network.

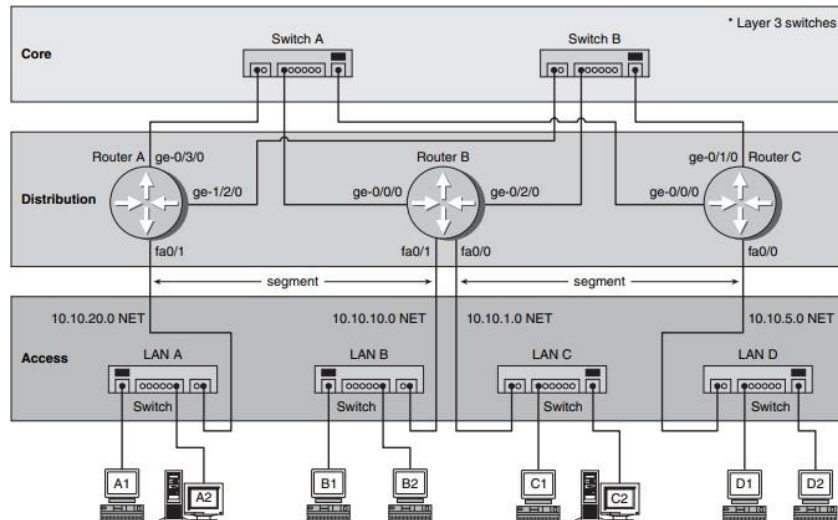
Campus Network Design

Core Layer

The network core usually contains high-end layer 3 switches or routers. The core is the heart or the backbone of the network. The major portion of a network's data traffic passes through the core. The core must be able to quickly forward data to other parts of the network. Data congestion should be avoided at the core if possible.

An example of a route policy is **traffic filtering**, which limits what traffic can pass from one part of a network to another. Key characteristics include:

- High speed and low latency.
- Redundancy for uninterrupted service during link failures. Minimal policy application to ensure efficient data flow



Distribution Layer

The distribution layer in the network is the point where the individual LANs connect to the campus network routers or layer 3 switches. Routing and filtering policies are more easily implemented at the distribution layer without having a negative impact on the performance of the network data traffic. Also, the speed of the network data connections at the distribution layer are typically slower than at the core.

Access Layer

The access layer is where the networking devices in a LAN connect together. The network hardware used here is typically a layer 2 switch. Hubs can be used but are not recommended in any networks with significant amounts of data traffic. Remember, a switch is a better choice because it forwards data packets directly to destination hosts connected to its ports.

Data Flow

An important networking issue is how data traffic flows in the core, distribution, and access layers of a campus LAN. In reference to Figure 9-1, if computer A1 in LAN A sends data to computer D1 in LAN D, the data is first sent through the switch in LAN A and then to Router A in the distribution layer. Router A then forwards the data to switch A or switch B. Switch A or switch B will then forward the data to Router C. The data packet is then sent to the destination host in LAN D.

Selecting the Media

The choices for the media used to interconnect networks in a campus network are based on several criteria:

- Desired data speed
- Distance for connections
- Budget

Load Balancing

Load balancing is the concept of distributing the network data traffic over multiple interfaces so that one interface connection is not overloaded. It is also used as a tool to control the direction of the data flow when multiple routes are available

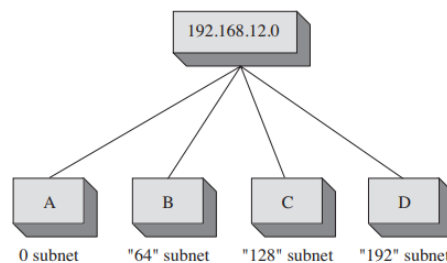
An example of load balancing is provided in the above Figure. In this case, data from LAN A going to LAN B will first pass through Router A at the distribution layer. The data will next travel to core Switches A or B. The data can all be sent to Switch A or Switch B or can be split over both switches.

IP ASSIGNMENT AND DHCP

IP assignment is a process where subnets are created for each subgroup or department. The IP address assignment is typically tracked by the network operations center (NOC). The IP addresses are kept in a central log file so that NOC can troubleshoot network problems.

IP addresses are assigned by NOC based on where the subnet for the computer is located. The subnet could be in a building, a floor of the building, a department, and so on. The subnets are created by the network administrators based on the expected number of users (hosts) in a subnet.

For example, the 192.168.12.0 network shown in below Figure has been partitioned into four subnets.



Subnet Addresses for the Subnets Shown in the above Figure

Subnet	Network Address	Broadcast Address
A	192.168.12.0	192.168.12.63
B	192.168.12.64	192.168.12.127
C	192.168.12.128	192.168.12.191
D	192.168.12.192	192.168.12.255

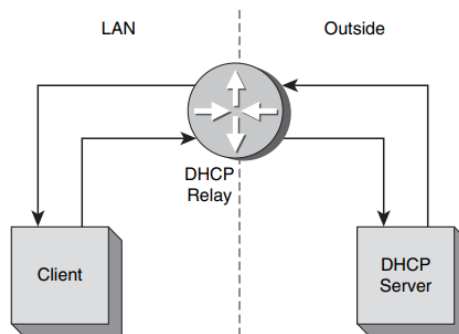
- **Static IP Assignment:**
 - Each device is manually assigned a unique IP address.
 - Suitable for servers, printers, and critical devices.
 - Time-consuming and prone to human errors.

- **Dynamic IP Assignment (DHCP):**

- Automates IP address allocation, managed by a DHCP server.
- Reduces administrative overhead and ensures efficient address utilization.

DHCP Configuration Steps:

1. Create a DHCP pool.
2. Define the network and subnet mask.
3. Specify the default gateway and DNS server.



An example requiring the use of a DHCP relay.

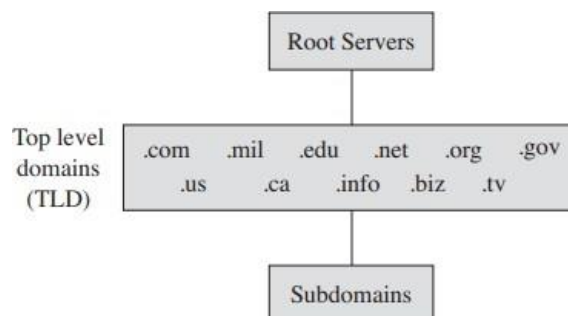
Network Services—DNS

DNS is the domain name service. DNS translates a human readable name to an IP address or an IP address to a domain name. The translation of a name to an IP address is called **forward domain name service**, and translation of an IP address to a domain name is called **reverse domain name service**.

Hierarchy of DNS:

1. **Root Servers:** Handle requests for top-level domains (e.g., .com, .edu).
2. **TLD Servers:** Manage domain-specific queries.
3. **Authoritative Servers:** Provide IP addresses for subdomains.

Example: To resolve `www.campus.edu`, the campus DNS server queries root servers, TLD servers, and authoritative servers, caching results for future use.



VLAN Configuration

Virtual LANs (VLANs) segment a network logically rather than physically, improving security and reducing broadcast traffic. Types include:

- **Port-Based VLANs:** Assign ports on a switch to specific VLANs.
- **Tag-Based VLANs (802.1Q):** Use VLAN IDs embedded in Ethernet frames for identification.
- **Protocol-Based VLANs:** Classify traffic based on protocols (e.g., IP, IPX).

Static VLAN Configuration Steps:

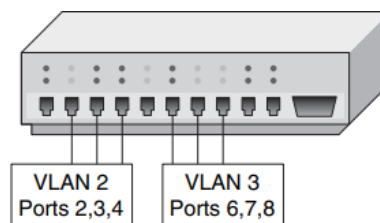
1. Access the switch's privileged mode.
2. Define VLANs and assign names.
3. Assign ports to the appropriate VLANs.
4. Verify configurations using commands like show vlan.

Example Commands:

Switch(config)# vlan 10

Switch(config-vlan)# name SALES

Switch(config-if)# switchport access vlan 10



Network Monitoring Using SNMP

Simple Network Management Protocol (SNMP) allows administrators to monitor and manage network devices by collecting performance data.

Components:

- **SNMP Manager:** Central console for monitoring.
- **SNMP Agent:** Resides on devices, responding to requests.
- **Management Information Base (MIB):** Database of managed objects.

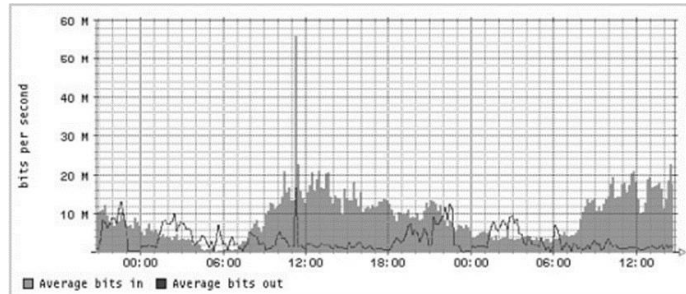
3.6 Data Traffic Analysis

Analyzing traffic helps identify usage patterns and potential issues. Tools like SNMP enable:

- **Hourly Monitoring:** Spotting traffic peaks.

- **Daily Trends:** Identifying recurring patterns.
- **Weekly/Monthly Reports:** Assessing long-term utilization.

Example: A campus administrator detects unusual spikes every Wednesday at noon. Using SNMP logs, they pinpoint the cause as scheduled software updates.



The hourly plot of a router's data traffic.

Conclusion

Configuring and managing a campus network requires a clear understanding of its hierarchical structure, efficient IP management, and robust monitoring systems. Utilizing VLANs, DNS, DHCP, and SNMP tools ensures optimal performance, scalability, and security. Regular traffic analysis aids in proactive troubleshooting and planning for future expansions.

1.4 NETWORK SECURITY

Network Security

Any action intended to safeguard the integrity and usefulness of your data and network is known as network security. In other words, Network security is defined as the activity created to protect the integrity of your network and data.

Working on Network Security

Network security uses several layers of protection, both at the edge of the network and within it. Each layer has rules and controls that determine who can access network resources.

The basic principle of network security is protecting huge stored data and networks in layers that ensure the bedding of rules and regulations that have to be acknowledged before performing any activity on the data. These levels are:

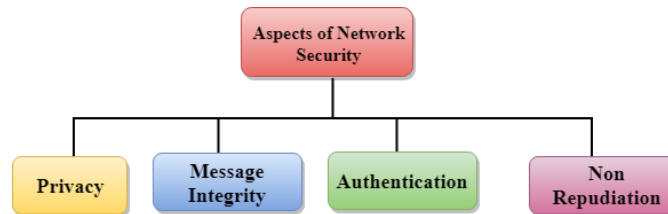
Physical Network Security: This is the most basic level that includes protecting the data and network through unauthorized personnel from acquiring control over the confidentiality of the network. The same can be achieved by using devices like biometric systems.

Technical Network Security: It primarily focuses on protecting the data stored in the network or data involved in transitions through the network. This type serves two purposes. One is protected from unauthorized users, and the other is protected from malicious activities.

Administrative Network Security: This level of network security protects user behavior like how the permission has been granted and how the authorization process takes place. This also ensures the level of sophistication the network might need for protecting it through all the attacks. This level also suggests necessary amendments that have to be done to the infrastructure.

Aspects of Network Security

Following are the desirable properties to achieve secure communication:



Privacy: Privacy means both the sender and the receiver expects confidentiality. The transmitted message should be sent only to the intended receiver while the message should be opaque for other users. Only the sender and receiver should be able to understand the transmitted message as eavesdroppers can intercept the message.

Message Integrity: Data integrity means that the data must arrive at the receiver exactly as it was sent. There must be no changes in the data content during transmission, either maliciously or accident, in a transit.

End-point authentication: Authentication means that the receiver is sure of the sender's identity, i.e., no imposter has sent the message.

Non-Repudiation: Non-Repudiation means that the receiver must be able to prove that the received message has come from a specific sender. The sender must not deny sending a message that he or she send.

Types of Network Security

There are several types of network security through which we can make our network more secure, Network and data are shielded from breaches, invasions, and other dangers by network security.

Email Security

Email Security is defined as the process designed to protect the Email Account and its contents safe from unauthorized access. For Example, you generally see, fraud emails are automatically sent to the Spam folder. Because most email service providers have built-in features to protect the content.

Network Segmentation

Network traffic is divided into several categories by software-defined segmentation, which also facilitates the enforcement of security regulations. Ideally, endpoint identity—rather than just IP addresses—is the basis for the classifications.

Access Control

Your network should not be accessible to every user. You need to identify every user and every device in order to keep out any attackers. You can then put your security policies into effect. Noncompliant endpoint devices might either have their access restricted or blocked.

Sandboxing

Sandboxing is a cybersecurity technique in which files are opened or code is performed on a host computer that simulates end-user operating environments in a secure, isolated environment.

Cloud Network Security

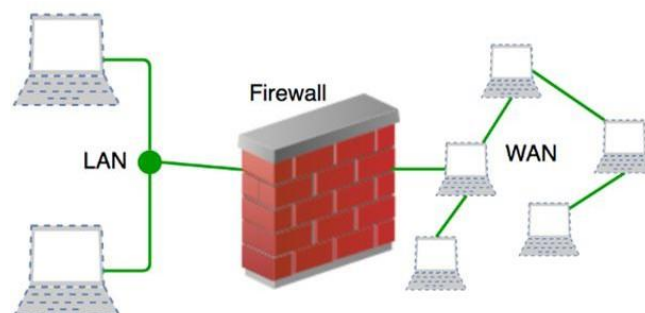
This is very vulnerable to the malpractices that few unauthorized dealers might pertain to. This data must be protected and it should be ensured that this protection is not jeopardized by anything.

Web Security

Online security solution will restrict access to harmful websites, stop web-based risks, and manage staff internet usage. Your web gateway will be safeguarded both locally and in the cloud.

Firewalls Security

A firewall is a network security device, either hardware or software-based, which monitors all incoming and outgoing traffic and based on a defined set of security rules accepts, rejects, or drops that specific traffic.



Application Security

Application security denotes the security precautionary measures utilized at the application level to prevent the stealing or capturing of data or code inside the application.

Web Security

A web security solution manages how your staff uses the internet, blocks threats from websites, and stops access to harmful sites. It safeguards your web gateway either onsite or in the cloud.

Mobile Device Security

Cybercriminals are focusing more on mobile devices and apps. In the next three years, about 90 percent of IT organizations might allow corporate applications on personal mobile devices.

Tools and Software for Network Security

Network to network, and with time, there are changes in the security tools and rules available. Strong security frequently requires various strategies, sometimes known as layered security or defense, to provide organizations with the most feasible security controls. The following are some examples of frequently used tools and software for network security:

1. Firewalls

Web pages, pop-ups, and other service entry and departure decisions are made by firewalls, which are guardian services or devices. Depending on the needs, these firewalls utilize a preset set of rules to help block or allow traffic. Depending on the requirements of the system, firewalls might be either software- or hardware-based, or both.

2. Access Control

Access control enables businesses to stop unauthorized people and devices from connecting to a specific network and to stop prospective attackers from accessing sensitive data. This limits network access to users who are authorized to utilize the specified resources.

3. Virtual Private Networks (VPN)

In most cases, a VPN encrypts the communication between an endpoint device and a network via the internet. Additionally, VPN enables experts to verify the connection between the network and the device. As a consequence, an online tunnel that is encrypted and safe is created.

4. Intrusion Prevention Systems

Intrusion prevention systems scan network traffic to identify and stop assaults. This is accomplished by connecting network activity with databases of attack methods that experts are familiar with.

5. Wireless Security

In comparison to wireless networks, wired networks could be more secure. It would help if you had control over the computers and people who may access the network of your business. It would help if you had wireless security, especially in light of the fact that fraudsters are increasingly extorting people for their private information.

6. Application Security

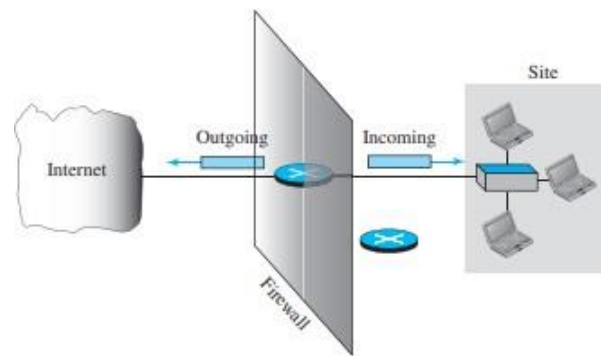
Applications' weak points may be tracked down and secured using a combination of software, hardware, and processes, which makes it harder for hackers to access your network.

7. Behavioural Analysis

You need to have a solid understanding of the typical behaviour of your network if you want to be able to spot abnormalities and different network breaches as they happen. Different behavioural analytics solutions are available that may quickly identify unusual activity.

1.5 FIREWALLS

A firewall is a device (usually a router or a computer) installed between the internal network of an organization and the rest of the Internet. It is designed to forward some packets and filter (not forward) others. The below Figure shows a firewall.



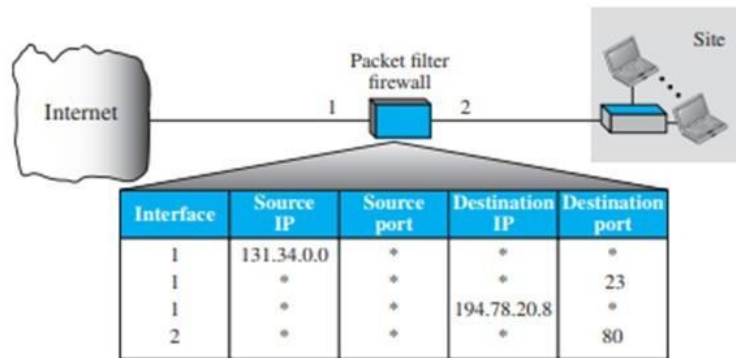
For example, a firewall may filter all incoming packets destined for a specific host or a specific server such as HTTP. A firewall can be used to deny access to a specific host or a specific service in the organization. A firewall is usually classified as a packet-filter firewall or a proxy-based firewall.

Packet-Filter Firewall

A firewall can be used as a packet filter. It can forward or block packets based on the information in the network-layer and transport-layer headers: source and destination IP addresses, source and destination port addresses, and type of protocol (TCP or UDP). A packet-filter firewall is a router that uses a filtering table to decide which packets must be discarded (not forwarded). The below Figure shows an example of a filtering table for this kind of a firewall.

According to the figure, the following packets are filtered:

1. Incoming packets from network 131.34.0.0 are blocked (security precaution). Note that the * (asterisk) means “any.”
2. Incoming packets destined for any internal TELNET server (port 23) are blocked.
3. Incoming packets destined for internal host 194.78.20.8 are blocked. The organization wants this host for internal use only.



4. Outgoing packets destined for an HTTP server (port 80) are blocked. The organization does not want employees to browse the Internet.

Proxy Firewall

The packet-filter firewall is based on the information available in the network layer and transport layer headers (IP and TCP/UDP). However, sometimes we need to filter a message based on the information available in the message itself (at the application layer).

As an example, assume that an organization wants to implement the following policies regarding its web pages: only those Internet users who have previously established business relations with the company can have access; access to other users must be blocked.

In this case, a packet-filter firewall is not feasible because it cannot distinguish between different packets arriving at TCP port 80 (HTTP). Testing must be done at the application level (using URLs).

One solution is to install a proxy firewall (computer) (sometimes called an application gateway), which stands between the customer computer and the corporation computer. When the user client process sends a message, the application gateway runs a server process to receive the request.

The server opens the packet at the application level and finds out if the request is legitimate. If it is, the server acts as a client process and sends the message to the real server in the corporation. If it is not, the message is dropped and an error message is sent to the external user. In this way, the requests of the external users are filtered based on the contents at the application layer.

